Horizontal component of Ear

50 Turns of coil (r = 6.8 cm)

Sl No.	Current (mA)	Deflection of Compass θ (Degree)		Deflection of compass by reversing current direction θ (Degree)		Average θ
		Left	Right	Left	Right	
1	30	19	19	20	20	19.5
2	40	24	24	25	25	24.5
3	54	31	31	32	32	31.5
4	62	35	35	36	36	35.5
5	76	40	40	41	41	40.5
6	85	43	43	45	45	44
7	113	50	50	53	53	51.5
8	137	55	55	58	58	56.5
9	167	60	60	64	64	62
10	226	66	66	70	70	68

500 Turns of coil (r = 6.8 cm)

Sl No.	Current (mA)	Deflection of Compass θ (Degree)		Deflection of compass by reversing current direction θ (Degree)		Average θ
		Left	Right	Left	Right	
1	3	18	18	18	18	18
2	4.2	24	24	24	24	24
3	5.6	30	30	30	30	30
4	6.9	36	36	36	36	36
5	8.8	42	42	43	43	42.5
6	10.4	47	47	49	49	48
7	13.1	53	53	55	55	54
8	16	58	58	60	60	59
9	19.3	62	62	65	65	63.5
10	25.4	67	67	71	71	69

th's magnetic field using Tangent galvanometer

θ(Radians)	tan θ	Bi = $\frac{\mu_0 N i}{2r}$ (Tesla)	$B_H = \frac{Bi}{tan\theta}$ (Tesla)	B_H(Micro Tesla)
0.3403	0.354	1.39E-05	3.91394E-05	39.13
0.4276	0.456	1.85E-05	4.05507E-05	40.55
0.5498	0.613	2.49E-05	4.07114E-05	40.71
0.6196	0.713	2.86E-05	4.01574E-05	40.15
0.7069	0.854	3.51E-05	4.11109E-05	41.11
0.7679	0.966	3.93E-05	4.06653E-05	40.66
0.8988	1.257	5.22E-05	4.15265E-05	41.52
0.9861	1.511	6.33E-05	4.18934E-05	41.89
1.0821	1.881	7.72E-05	4.10235E-05	41.02
1.1868	2.475	0.000104	4.21852E-05	42.18

θ(Radians)	tan θ	Bi = $\frac{\mu_0 N i}{2r}$ (Tesla)	$B_H = \frac{Bi}{tan\theta}$ (Tesla)	B_H(Micro Tesla)
0.3142	0.325	1.39E-05	4.26567E-05	42.65
0.4189	0.445	1.94E-05	4.35821E-05	43.58
0.5236	0.577	2.59E-05	4.48116E-05	44.81
0.6283	0.727	3.19E-05	4.38763E-05	43.87
0.7418	0.916	4.07E-05	4.43682E-05	44.36
0.8378	1.111	4.8E-05	4.32626E-05	43.26
0.9425	1.376	6.05E-05	4.39718E-05	43.97
1.0297	1.664	7.39E-05	4.44156E-05	44.41
1.1083	2.006	8.92E-05	4.44565E-05	44.45
1.2043	2.605	0.000117	4.50457E-05	45.04